

SurvFormer: A Transformer Based Framework for Survival Analysis in Insurance Underwriting

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Abstract

Insurance claim prediction is a critical task in the underwriting process of health insurance. It helps keep insurance programs financially stable and ensures that people have fair access to coverage. In industrial insurance systems, predictive modeling based on a large amount of user data accumulated from online insurance platforms can significantly improve underwriting accuracy. However, large amounts of policies are newly underwritten especially for some newly developed insurance products. The time lag between a policy being underwritten and a corresponding insurance claim can range from several months to years. Thus, newly acquired data may lack accurate label information when used by the traditional classification algorithm, presenting a data censoring challenge in the prediction of insurance claims. Moreover, the timing of insurance claims can be significantly influenced by distinct features associated with the policyholder's behavior and circumstances. In addressing this problem, we incorporate survival analysis into our modeling framework. We propose SurvFormer, an innovative transformer-decoder-based approach designed to estimate temporal survival probabilities. SurvFormer utilizes attention mechanisms to highlight relevant user behaviors and effectively captures interdependencies among temporal survival states. Furthermore, to address the prevalent issue of event-censoring imbalance in survival

data, we design a survival loss called imbalance-aware censoring loss (IACL) to prevent the model from being overwhelmed by the prevalence of samples that lack precise labels. Through comprehensive experiments on three real-world data sets, SurvFormer achieves significant performance improvement and maintains high efficiency compared to the state-of-the-art baseline methods.

CCS Concepts

• Information systems → Data mining.

Keywords

Insurance Underwriting, Survival Analysis, Transformer, Data Imbalance

ACM Reference Format:

Yingxue Li, Xiaolu Zhang, Jun Hu, Wenwen Xia, Ziyi Liu, Xiaobo Qin, Binjie Fei, and Jun Zhou. 2025. SurvFormer: A Transformer Based Framework for Survival Analysis in Insurance Underwriting. In *Companion Proceedings of the ACM Web Conference 2025 (WWW Companion '25)*, April 28-May 2, 2025, Sydney, NSW, Australia. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3701716.3715262>

1 Introduction

In health insurance, policyholders file claims to get reimbursed by their insurer for covered medical consultations, tests, or treatments. Therefore, in the context of purchasing an insurance policy, the underwriting process plays a pivotal role, where insurers evaluate both current and future risks to set premiums that accurately reflect the likelihood and potential cost of these claims. As the online insurance industry continues to evolve, advancements in data collection have enabled insurers to harness vast amounts of user behavior data, particularly from online insurance platforms. This data-rich environment presents an opportunity to enhance the precision of underwriting processes, ultimately leading to more

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WWW Companion '25, April 28-May 2, 2025, Sydney, NSW, Australia

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ACM ISBN 979-8-4007-1331-6/25/04

<https://doi.org/10.1145/3701716.3715262>

effective risk management and pricing strategies. However, three major challenges exist in the field.

(1) *Lack of accurate label information.* As the rapid introduction of new products and policies in the insurance industry, the absence of accurate label of the policies poses significant challenges to the effective training of traditional classification models. For example, as illustrated in Figure 1(a), when considering one-year medical insurance applications, insurers must predict whether a client will file a claim within the next year. However, the training data often includes policies that last less than a year without any claims. These policies are devoid of definitive binary outcomes that indicate a claim or no claim, making it difficult to classify these instances using traditional binary classification methods without introducing bias [1, 27]. We refer to these samples as censored in our study [3].

(2) *Varied patterns of claim.* The timing of insurance claims can be significantly influenced by distinct features associated with the policyholder's behavior and circumstances. For example, as illustrated in Figure 1(b), the top policyholder recently visited the hospital and purchased medications through an online drug platform one month prior to purchasing the insurance policy. This individual has a high probability of filing a claim in the near future, as they are likely already aware of their health issues and have bought the policy specifically for reimbursement purposes. Conversely, the bottom policyholder files a claim after a relatively longer period, potentially indicating a lifestyle-related natural risk.

(3) *Data imbalance.* In the context of health insurance, censored samples constitute a significant majority, even exceeding 99% of the dataset. This imbalance issue can skew models towards biased predictions due to the overwhelming influence of censored samples.

For the first issue, survival analysis techniques [3] are suitable to address the challenge by incorporating the time-to-event information (Figure 1(c)) into the modeling process. By integrating time-to-event data, survival analysis enriches the training dataset, enabling a deeper understanding of event likelihood and timing, which is vital for precision in underwriting decisions. Previous survival analysis techniques can broadly be classified into three primary groups. The first and second categories encompass statistical methods [4, 5, 12] and machine learning approaches [2, 17, 24] respectively, both of which typically involve rigid assumptions regarding the forms of survival probabilities. The third category consists of deep learning models, exemplified by DeepHit [22] and DRSA [31]. These deep learning approaches either overlook the dependencies between survival states or are not well-suited for large-scale industrial datasets due to limited computational efficiency.

For the second issue, the varied claim patterns present unique challenges within the health insurance field. To the best of our knowledge, there are no survival techniques explicitly designed to address this issue.

For the third issue, in the classification domain, the issue of class imbalance has prompted the development of several tailored loss functions. However, these functions fall short when dealing with censored data. Consequently, there is an imperative need to devise methodologies capable of addressing class imbalance in datasets that contain censored instances.

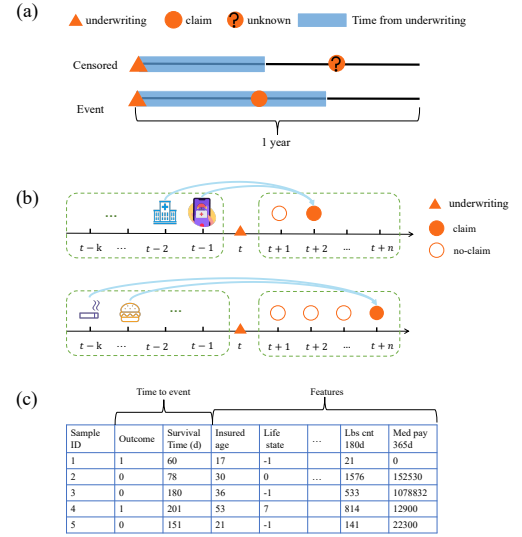


Figure 1: Illustrative examples in health insurance

To address the aforementioned challenges, this paper introduces SurvFormer, an innovative approach utilizing a transformer-decoder architecture to effectively model temporal survival probabilities. Additionally, we propose a novel survival loss function to mitigate data imbalance issues. SurvFormer adapts to the insurance claim scenario and overcomes the limitations of previous methodologies in the following ways.

1) *Attention mechanism.* Firstly, by employing a cross-attention mechanism, SurvFormer is capable of focusing on various parts of the features when predicting future survival probabilities over time. Secondly, the self-attention component within SurvFormer is designed to thoroughly understand the inter-dependencies between past and current survival states across the prediction timeline. This mechanism allows the model to capture temporal dynamics and their influence on present conditions more effectively. Furthermore, the attention mechanism's capability for parallel sequence processing significantly enhances the practicality of handling extensive industrial datasets.

2) *Imbalance-Aware Censoring Loss (IACL)* We design a survival loss specially by introducing modulating factors to reduce the influence of easy, predominantly censored samples and concentrate on difficult, mainly event samples. The novel loss reduces the overwhelming influence of censored samples and targets critical prediction probabilities more effectively.

In conclusion, the salient contributions of this work are outlined as follows:

- To dynamically focus on relevant features and the previous survival states, the cross-attention and self-attention are used to grasp the complex patterns in survival data, leading to improved predictions. Furthermore, the parallel sequence processing capability of our method is well-suited to handling the large-scale datasets commonly encountered in industrial applications.

- To address the prevalent challenge of event-censoring imbalance, we propose an innovative survival loss function called Imbalance-Aware Censoring Loss (IACL). This function significantly enhances the model's discriminative power.
- The effectiveness of SurvFormer is validated on one industrial insurance dataset and two public medical datasets. Comparative experiments demonstrate that SurvFormer achieves superior performance against existing state-of-the-art methodologies.

2 Related Work

The related work intersects two core areas: survival analysis and the sequence modeling. Each of these domains contributes a unique aspect to our proposed methodology.

2.1 Survival Analysis

Survival analysis is a critical field of statistics that focuses on predicting the time until an event of interest occurs particularly in the presence of censored data. Previous survival analysis techniques can broadly be classified into three primary groups. The statistical category, represented by models such as the Cox proportional hazards model [5], the Weibull survival model [4], and the exponential survival model [12], typically make rigid assumptions on the hazard or survival function. These models also presuppose a linear association between the features and the survival or hazard functions.

The second category is the machine learning branch, typified by models like the random survival forest [17] and the survival variation of XGBoost [2, 24]. Although these models loosen the assumptions on the impact of features on the hazard function, they still maintain a proportional hazard speculation.

The third category is composed by deep learning models. DeepHit [22] is designed to directly estimate the probabilities of events at predetermined time points without presupposing the survival distribution. However, it overlooks the sequential dependencies of the survival probabilities between time points. In contrast, DRSA [31] employs the recurrent neural network (RNN) to capture the temporal dependencies through modeling the hazard function over the discrete time space. However, it may encounter long-term dependency issues as the prediction window extends. Moreover, when implemented on extensive industrial data, the efficiency is also a matter of concern, primarily due to its sequential training scheme. DSM [30] estimates an average of mixed parametric distributions based on deep learning framework, which imposes certain assumptions on survival distributions through its choice of parametric distributions. The Implicit Survival Function (ISF) [25] employs Implicit Neural Representations to forecast the hazard function across temporal dimensions. Nonetheless, it utilizes time encoding for differentiating how features affect survival probabilities at distinct time intervals, which falls short in capturing the complex relationship among the features and the survival probabilities.

In the context of health insurance, the data exhibits distinct patterns and poses unique challenges, rendering the direct application of existing survival analysis methodologies nontrivial. To address these complexities, our research is specifically tailored to handle censored data within the health insurance sector.

2.2 Sequence modeling

The transformer [33] has revolutionized the field of natural language processing (NLP) with its innovative self-attention mechanism, which captures long-range dependencies across input sequences. Its ability to process sequences in parallel and learn contextual representations has made it surpass traditional recurrent neural networks (RNNs) [10] and long short-term memory (LSTM) networks [16]. This architecture has proven to be highly effective in a variety of tasks, such as machine translation [33], text summarization [26], language understanding [34], question answering [11], and language generation [15]. Outside the scope of natural language processing, the transformer architecture has also been successfully implemented, including computer vision [13], speech recognition [8], and bioinformatics [32]. The flexibility, scalability and proficiency in learning contextual representations of the transformer architecture make it a prime candidate for adaptation in survival models.

3 Methods

3.1 Preliminaries in Time-to-event Data

In our study, we focus on the concept of right-censoring, which is the predominant circumstance encountered in survival analysis [5, 18]. We represent the survival data set with the notation $D = (X^{(i)}, t^{(i)}, z^{(i)})_{i=1}^n$, where i indexes an individual subject, $X^{(i)}$ denotes their feature vector, and n is the total number of samples. For each subject i , $t^{(i)}$ refers to the event time for event samples or the last time for censored samples. The binary indicator $z^{(i)} \in \{0, 1\}$ signifies whether the event of interest has occurred within the observed time. Specifically, a value of 0 indicates that the event has not occurred by the end of the observation window (censored), and 1 indicates the event occurred at time $t^{(i)}$. In our study, we model the survival distribution in the discrete time space as in DeepHit [22] and DRSA [31]. The time space is uniformly divided into L disjoint time slices $0 < t_1 < t_2 < \dots < t_L$. Each interval is represented as $V_l = (t_{l-1}, t_l]$, with V_L signifying the final interval in the entire data range. Consequently, the continuous time frame $(0, t_L]$ is segmented into a series of discrete intervals $\{V_0, V_1, \dots, V_L\}$.

3.2 Survival function and hazard function

The survival function (survival rate) and hazard function (hazard rate) are two fundamental functions in survival analysis [9]. In this section, we begin by presenting their formulations within the context of continuous time and subsequently adapt these concepts to the discrete-time framework.

3.2.1 Continuous time domain. In standard survival analysis, the survival function denotes probability of surviving beyond time t [23], which is mathematically defined as:

$$S(t) = \Pr(T > t) = 1 - F(t) = 1 - \int_0^t f(u) du \quad (1)$$

where T is a continuous random variable indicating the event occurred time. $F(t) = \Pr(T < t)$ denotes the cumulative distribution function (c.d.f) presenting the probability that the event has occurred by time t . Correspondingly, $f(t)$ signifies the probability

density function (p.d.f), which describes the probability of the event occurring at an infinitesimal interval around time t .

The hazard function, denoted as $h(t)$, represents the instantaneous rate of occurrence for an event at time t , conditional on the event not having occurred prior to this time [9]. Mathematically, the hazard function is defined as:

$$\begin{aligned} h(t) &= \lim_{\Delta t \rightarrow 0} \frac{Pr(t < T \leq t + \Delta t | T \geq t)}{\Delta t} \\ &= \frac{f(t)}{S(t)} = \frac{f(t)}{1 - F(t)} \\ &= -\frac{\partial}{\partial t} \log(1 - F(t)) = -\frac{\partial}{\partial t} \log(S(t)) \end{aligned} \quad (2)$$

Then $S(t)$ can be expressed in terms of $h(t)$ as follows:

$$S(t) = \exp\left(-\int_0^t h(u) du\right) \quad (3)$$

3.2.2 Discrete time domain. Consequently, in the discrete-time domain, the formulations for survival function and hazard function are as follows:

$$S(l) = Pr(T > t_l) = \sum_{j>l} Pr(T \in V_j) \quad (4)$$

The discrete conditional hazard rate h_l can be defined as

$$\begin{aligned} h_l &= Pr(T \in V_l | T > t_{l-1}) \\ &= \frac{Pr(T \in V_l)}{Pr(T > t_{l-1})} \\ &= \frac{S(l-1) - S(l)}{S(l-1)} \end{aligned} \quad (5)$$

Then $S(l)$ and $f(l)$ can be determined from $h(l)$ by applying the probability chain rule as follows:

$$S(l) = S(l-1)(1 - h_l) = \prod_{j \leq l} (1 - h_j) \quad (6)$$

$$p(l) = Pr(T \in V_l) = S(l-1) - S(l) = h_l \prod_{j \leq (l-1)} (1 - h_j) \quad (7)$$

3.3 SurvFormer Framework

After demonstrating the survival function and hazard function in the discrete time space, we will describe the proposed SurvFormer model in detail. SurvFormer is architected with a transformer decoder coupled with a Multi-Layer Perceptron (MLP) encoding network. The comprehensive layout of the SurvFormer is showed in Figure 2.

The MLP encoding substructure is designed to derive the preliminary embeddings from input features, serving as the input for the transformer decoder. As depicted on the left side of Figure 2, the input features for each instance are categorized into categorical and continuous types. The continuous features are first processed through a fully connected (FC) layer with dropout to obtain a dense embedding. The numbers in parentheses indicate the quantity of units. Categorical features are one-hot encoded first and then converted into dense tensors a sparse embedding layer. Subsequently, these dense representations are concatenated and

processed through several repeated FC layers with dropout. The output of the encoder block, denoted as $f_{enc}(x)$, carries vital information predicting the future survival status and is subsequently fed into a transformer decoder.

The transformer decoder was employed to estimate the hazard function h_l within each time interval V_l . The detailed structure was illustrated in the right side of Figure 2. When predicting h_l within time interval V_l , the decoder utilizes the encoder's output $f_{enc}(x)$, combined with the corresponding time stamp $\{T_1, T_2, \dots, T_{l-1}\}$, as its input. The time stamp T_i , which matches the left boundary of the interval V_i , is utilized as positional encoding to convey temporal information associated with the target survival states. This encoding is implemented using predefined sinusoidal functions formalized in Equation (8). d_{model} is the dimension of the representations used as input to the multi-head attention.

$$\begin{cases} PE(t, 2i) = \sin(t/1000^{2i/d_{model}}) \\ PE(t, 2i+1) = \cos(t/1000^{2i/d_{model}}) \end{cases} \quad (8)$$

The masked self-attention in the transformer decoder allows each position (time interval) to attend to all previous positions in the previous layer of the decoder, which provides a rich and contextual understanding of the survival states and the order of the states being generated. The cross-attention mechanism takes the query vectors of the target survival states from the current decoder layer and the key and value vectors from the encoder output $f_{enc}(x)$. It computes attention scores that indicate how much focus should be placed on each part of the encoder output when generating the next h_l in the decoder output. These scores are then used to create a weighted sum of the value vectors, producing an output that is passed through the subsequent layers of the decoder. Then the output of the last decoder block is transformed into a predicted hazard probability distribution using a linear layer followed by a softmax activation.

Finally, the output $H = \{h_0, h_1, \dots, h_l\}$ is transformed into the survival probability distribution $S = \{S_0, S_1, \dots, S_l\}$, $p = \{p_0, p_1, \dots, p_l\}$ using Equation (6) and Equation (7) for loss calculation.

3.4 Imbalance-Aware Censoring Loss (IACL)

We design a survival loss to handle the censored/event imbalance problem in the survival data. Specifically, our loss function is composed of three components: loss for event samples, loss for censored samples, and a modulating factor to handle data imbalance.

3.4.1 loss for event samples. The first loss aims to minimize the negative log-likelihood of the true event time for the event subject, which is mathematically expressed as

$$\begin{aligned} L_{event}^{(i)} &= -\log p(l^{(i)}) \\ &= -\log Pr(T \in V_{l^{(i)}} | x^{(i)}, \theta) \\ &= -\log[h_{l^{(i)}}^{(i)} \prod_{j \leq l^{(i)}-1} (1 - h_j^{(i)})] \\ &= -[\log h_{l^{(i)}}^{(i)} + \sum_{j \leq l^{(i)}-1} \log(1 - h_j^{(i)})] \end{aligned} \quad (9)$$

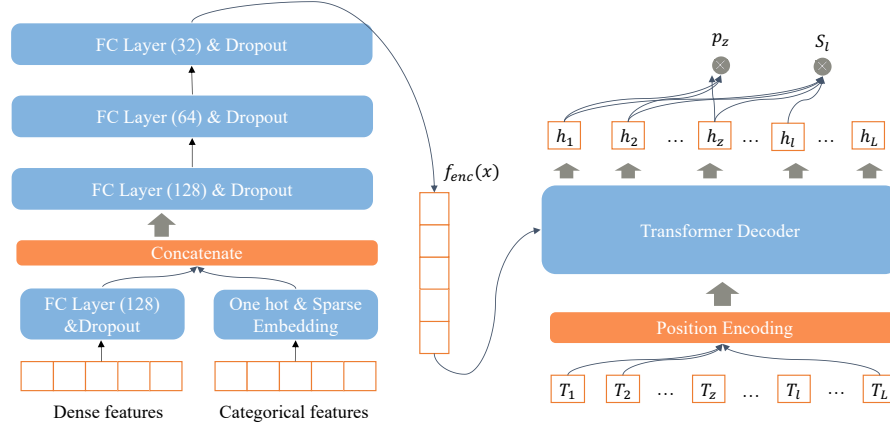


Figure 2: The architecture of SurvFormer

3.4.2 loss for censored samples. The second loss function aims to maximize the survival probability up to the last observed time point, as no event has occurred during the follow-up period.

$$\begin{aligned}
 L_{censored}^{(i)} &= -\log S(l^{(i)} | x^{(i)}; \theta) \\
 &= -\log \prod_{j \leq l^{(i)}} (1 - h_j^{(i)}) \\
 &= \sum_{j \leq l^{(i)}} \log(1 - h_j^{(i)})
 \end{aligned} \quad (10)$$

3.4.3 Modulating factor for imbalanced event-censoring Data. As in many survival data set, the event and censored samples are extremely imbalanced, where the event samples take even less than 1% percentage. This can lead to a skew in the loss function, where censored samples dominate the loss, resulting in poor model performance on non-censored samples. Inspired by focal loss, we introduce modulating factors to reduce the influence of easy, predominantly censored samples and concentrate on difficult, mainly event samples. The modulated loss for the censored/event samples are formulated as follows:

$$RFL_{censored/event}^{(i)} = (1 - q^{(i)})^\gamma L_{censored/event}^{(i)} \quad (11)$$

The term $(1 - q^{(i)})^\gamma$ is the modulating factor which reduces the loss contribution from easy samples and increases the contribution from hard examples. As $q^{(i)}$ approaches 1, the modulating factor goes to 0 for easy examples, reducing their influence on the loss. In contrast, when $q^{(i)}$ is quite small, the modulating factor approaches 1, leaving the loss unaffected. For the event sample i , we hope the event probability in the time interval $V_l^{(i)}$ approaches 1. When the predicted probability is small, the sample should be assigned larger weight. While for the censored sample, we hope its survival probability at time t_i should approaches 1. Thus, we define $q^{(i)}$ as

$$q^i = \begin{cases} \Pr(T^{(i)} \in V_l^{(i)} | x^{(i)}, \theta), & i \in event \\ S(t^{(i)} | x^{(i)}; \theta), & i \in censored \end{cases} \quad (12)$$

The parameter γ serves as the focusing factor, a positive scalar that reduces the loss contribution from well-classified examples and shifts the emphasis towards difficult, misclassified ones. As γ increases, well-classified or "easy" examples receive less weight in the loss calculation. With When γ equals 0, the loss remains unchanged.

The total loss is the combination of the losses for event and censored samples accompanied with the modulated factor as follows:

$$L = \frac{1}{N} \left(\sum_{i \in event} RFL_{event}^{(i)} + \sum_{i \in censored} RFL_{censored}^{(i)} \right) \quad (13)$$

4 Experiment

4.1 Datasets

The performance of SurvFormer was evaluated by comparing it with several state-of-art methods on three real-world data sets, including one insurance dataset and two clinical data sets. Brief summaries of these data sets are provided below. The statistics of the three datasets is shown in Table 1.

- **INSURANCE** The industrial insurance dataset is comprised of 17.4 million medical insurance policies from Company A, underwritten from January 2020 to June 2023. Each policy in the dataset is accompanied by 151 features that span demographic information, lifestyle habits, healthcare spending patterns, outpatient and inpatient engagements, medical history, and previous insurance actions, etc. For policies with claims, the time from underwriting to the claim is recorded. For policies without claims, the length of time the policy has been in force is noted.
- **SUPPORT** The Study to Understand Prognoses Preferences Outcomes and Risks of Treatment (SUPPORT) investigation is an extensive study that examines the survival durations of critically ill hospitalized adults [20]. This data set includes data on 9,105 patients across 14 variables, with nearly complete records for each patient (including age, sex, race, comorbidity count, diabetes status, dementia status, cancer status, mean arterial pressure, heart rate, respiratory rate, body temperature, white blood cell count, serum sodium

Table 1: Descriptive statistics of INSURANCE, SUPPORT and FLCHAIN

Dataset	#Total Samples	#Censored Samples	Censored rate	#Features	Max Days
INSURANCE	12,045,534	111,635	0.991	151	366
SUPPORT	8,873	2,837	0.320	14	85
FLCHAIN	6,524	4,562	0.699	8	5,166

levels and serum creatinine levels). Additionally, it logs the duration from enrollment to the date of death or the most recent date of contact for surviving participants. Patients with missing variables were excluded, leaving a final count of 8,873.

- **FLCHAIN** The Assay of Serum Free Light Chain (FLCHAIN) dataset contains 6,524 individuals from a study exploring the relationship between serum free light chain (FLC) and mortality[7, 21]. The dataset includes various variables including age, sex, year of blood sample collection, kappa and lambda portions of serum free light chain, FLC group classification, serum creatinine levels, and monoclonal gammopathy of undetermined significance (MGUS) status. It also records the time from enrollment to either the date of death or the last known contact date for surviving samples.

4.2 Baseline Methods

- **Cox.** The Cox proportional hazards model [5] is a semi-parametric model widely used in survival analysis. It assumes that the hazard function for an individual is a product of a baseline hazard function and an exponential function of co-variables. The key feature of the Cox model is its assumption of proportional hazards, meaning that the hazard ratios between individuals are constant over time.
- **XGB-sur.** XGBoost is a powerful ensemble learning method based on boosted decision trees, which has been adapted for survival analysis [2, 24]. The survival version of XGBoost typically involves modifications to the objective function to handle censored data appropriately. We employ the cox loss when training XGB-sur model.
- **DeepHit.** DeepHit [22] is a deep learning-based model designed for survival analysis. It uses a neural network to directly predict the probability distribution of survival times, capturing complex interactions among covariates. It provides a flexible framework for learning survival distributions without explicit assumptions.
- **DRSA.** Deep Recurrent Survival Analysis (DRSA) [31] leverages recurrent neural networks for survival analysis. DRSA predicts the time-to-event outcomes by learning complex temporal dependencies, allowing for improved handling of time-varying risk factors compared to traditional survival models.

The reproduction of DeepHit and DRSA is based on the official code ^{1 2}.

¹<https://github.com/rk2900/DRSA>

²<https://github.com/havakv/pycox>

4.3 Evaluation Metric

We utilize the time-dependent concordance index (C^{td} -index) [14] to assess the performance of the survival models, a metric commonly adopted in survival analysis [6, 29]. The C^{td} -index is a generalization of the area under the ROC curve (AUC) [28], which can take into account censored data. It provides a global assessment of the model's discrimination power by correctly ranking individual survival times based on their estimated risk scores. It can be computed with the following formula:

$$A_{i,j} \triangleq \mathbb{1}(T_i < T_j) \quad (14)$$

$$C^{td} = \frac{\sum_{i \neq j} A_{i,j} * \mathbb{1}(S(l_i|x_i) < S(l_i|x_j))}{\sum_{i \neq j} A_{i,j}} \quad (15)$$

$A_{i,j}$ is the indicator function of comparable pairs (i, j) . To be noted, T_i, T_j are true event time for sample i and j , which can be obtained only for event samples. Thus, the sample pair is comparable either when both of the two samples are event samples or the observation time for the censored sample are beyond the event time of the event sample. Two censored samples cannot be involved into the pairwise comparison as their actual event times are unknown. $S(l_i|x_i)$ and $S(l_i|x_j)$ are the predicted survival rates for sample i and j at the time interval l_i as defined in Equation (6). Thus, the C^{td} estimates the percentage of correctly ranked pairs among all comparable pairs.

4.4 Experiment Setting

Throughout the evaluations, we take 30 days as the basic time interval for the INSURANCE dataset, 1 day for the SUPPORT dataset, and 30 days for the FLCHAIN dataset, respectively. For all experiments, the number of hidden units of the FC layer of the MLP encoding substructure is set as shown in Figure 2 and the drop out rate is 0.5. In the transformer decoder substructure, the dimension of the input representations for the multi-head attention, denoted as d_{model} , is set to 64. Meanwhile, The dimension of the feed-forward layers d_{ff} is set to 64.

During training, we perform Adam optimizer [19]. Models of the best C^{td} -index are selected by varying hyper-parameters of learning rate $\{10^{-3}, 10^{-4}, 10^{-5}\}$, batch size $\{128, 256, 512, 1024\}$, the number of decoder layers $\{2, 4, 6\}$ and the number of attention heads $\{2, 4, 6\}$. The influence of γ will be discussed in the ablation study.

For the INSURANCE dataset, the training and testing set were split according to the underwriting date. Policies underwritten before December 2021 formed the training set, and those underwritten after that date comprised the test set. This approach helps the offline evaluation better predict the model's online performance. As

the testing set being fixed, we performed five-fold cross-validation on the training set to select the optimal hyper-parameters. Subsequently, we evaluated the model performance on the independent test set. For the SUPPORT and FLCHAIN datasets, five-fold cross-validation were performed.

4.5 Performance Comparison

Table 2 displays the discriminative capabilities of various models as measured by the $C^{td} - index$. The mean and standard deviation from the five-fold cross-validation experiments are calculated and listed in the table. The findings from the table can be summarized as follows:

Firstly, the models that adopt predefined assumptions about the distribution of event times, such as the Cox proportional hazards model, XGBoost, and DeepSurv, tend to exhibit less favorable $C^{td} - index$ scores. The likely cause of this sub-optimal performance is that such rigid assumptions may not hold true in the face of complex, real-world data. Consequently, the inflexibility of these models to adapt to the actual distribution of event times limits their effectiveness in accurately predicting these outcomes.

Secondly, models like DeepHit, DRSA, and SurvFormer which are free of rigid distributional assumptions regarding the time-to-event data confers a significant advantage. Their distribution-agnostic characteristics allow them to more adeptly capture the nuances and heterogeneity present within the dataset. This flexibility enables these models to construct a more accurate and individualized representation of the survival probabilities for each subject, resulting in enhanced predictive performance.

Finally, our SurvFormer model distinguishes itself by leveraging the innovative cross-attention and masked self-attention mechanisms. These attention-based techniques allow SurvFormer to focus on the most relevant features and the previous survival states, which enables the model to estimate the survival probability distribution with greater precision. The incorporation of the modulating factor also significantly contributes to the performance on the INSURANCE dataset, which will be examined thoroughly in the forthcoming ablation study section.

Table 2: Comparison of C^{td} -index across three datasets: INSURANCE, SUPPORT and FLCHAIN

Method	C^{td} -index		
	INSURANCE	SUPPORT	FLCHAIN
Cox	0.556 (0.002)	0.569 (0.013)	0.756 (0.056)
XGB-sur	0.587 (0.005)	0.581 (0.010)	0.763 (0.057)
DeepSurv	0.600 (0.001)	0.695 (0.010)	0.768 (0.094)
DeepHit	0.604 (0.005)	0.719 (0.057)	0.774 (0.012)
DRSA	0.607 (0.005)	0.722 (0.052)	0.776 (0.004)
SurvFormer	0.614 (0.001)	0.730 (0.002)	0.789 (0.005)

4.6 Attention visualization

To gain a deeper and more intuitive understanding of the varied patterns of claims, as illustrated in Figure 1(b), we visualize the cross-attention matrices from the first and last (4th) layers. We specifically averaged the cross-attention matrices for samples who filed claims 1 month and 6 months after underwriting, respectively. For samples with claims at the 1st month, we visualized their cross-attention weights at the first time step on the encoder’s output embeddings, whereas for the 6th month, we visualized the weights at the sixth time step. In Figure 3, the x-axis denotes the indices of the embedding dimensions, while the y-axis represents the two groups of samples.

From Figure 3, we observe that the model generally learns to assign different weights to different parts of the encoder’s output. This observation confirms the varied patterns of the claims and underscores the substantial performance improvement achieved by our SurvFormer model.

4.7 Computational Efficiency

We evaluated the performance of our proposed model, SurvFormer, against traditional deep learning baselines in terms of runtime efficiency on the INSURANCE dataset, as shown in Table 3. The experiments were conducted using 16 worker nodes, each equipped with 16 CPU cores and 10 GB of memory, with a batch size of 1024 over 3 epochs. The findings demonstrate that SurvFormer is more efficient compared to the RNN-based DRSA method, primarily due to its ability to leverage parallelism through the attention mechanism. Moreover, the inference phase of SurvFormer is also faster than DRSA. This is because our model only requires the time step as input, thus eliminating the need for sequential dependencies on previous outputs, as illustrated in Figure 2. Compared to DeepSurv and DeepHit, SurvFormer has only a marginal increase in both training and inference time but achieves significantly better performance as shown in Table 2.

Table 3: Time cost of SurvFormer and the baseline methods.

Methods	Training (s)	Inference (s)
DeepSurv	4,101	431
DeepHit	5,017	445
DRSA	16,505	1,573
SurvFormer	5,495	496

4.8 Scaling effect

The scaling effect is a critical factor to consider when integrating more data and increasing model sizes, particularly in industrial applications. We conducted a comprehensive evaluation of SurvFormer’s performance under various data volumes and model sizes, as detailed in Table 4. The configurations for the number of decoder layers and attention heads were tested in combinations of (2, 4) and (2, 4, 6), respectively. Our experiments involved three datasets: INSURANCE (12 million samples, 151 features), SUPPORT (8.9 thousand samples, 14 features), and FLCHAIN (6.5 thousand samples, 8 features).

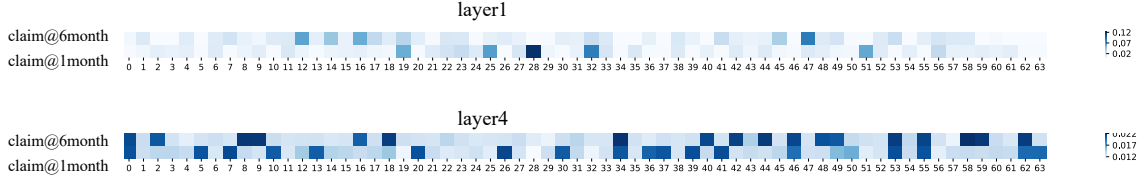


Figure 3: Cross-attention weights for samples who filed claims at different time points

The results indicate that the model’s scalability is particularly evident with larger datasets, such as the INSURANCE dataset. As SurvFormer’s parameters are increased, we observe an improvement in the C^{td} -index. For the SUPPORT dataset, although there is a noticeable performance enhancement, it is relatively moderate, tapering off at 2 layers and 6 attention heads. In the case of the smaller FLCHAIN dataset, excessively large models tend to overfit, leading to diminished performance. These findings underscore the importance of tailoring model size to the dataset at hand to achieve optimal results.

Table 4: C^{td} -index of SurvFormer on three datasets with varying numbers of decoder layers and attention heads

(#layers,#heads)	INSURANCE	SUPPORT	FLCHAIN
(2, 2)	0.604	0.728	0.789
(2, 4)	0.605	0.729	0.787
(2, 6)	0.609	0.730	0.785
(4, 2)	0.612	0.724	0.780
(4, 4)	0.614	0.723	0.779
(4, 6)	0.614	0.723	0.778

4.9 Ablation Study

In this ablation study, we conduct experiments to identify how much each part of the model contributes to its overall performance.

Firstly, to evaluate the benefit from the proposed imbalance-aware censoring loss (IACL), we varied the hyper-parameter $\gamma \in \{0, 0.5, 1, 2, 5\}$ in Equation (11), which controls the strength of the modulating factor. The experiments were conducted on the INSURANCE data set because it exhibits the most notable imbalance between event and censored samples. As shown in Table 5, when $\gamma = 0$, our loss is equivalent to the original survival loss without modulating factor. As γ increases, the “easy” samples with low loss get further discounted and the model pay more attention to the “hard” samples. Thus, the model performance increases from 0.610 to 0.614 as γ increases from 0 to 2. When γ reaches 5, the C^{td} -index decreases slightly mostly because the model paid over attention to the hard samples.

Secondly, to rigorously assess the efficacy of the transformer decoder architecture, we employ the identical IACL loss for both DRSA and DeepHit, analogous to its application in SurvFormer. As illustrated in Table 6, the C^{td} - index for both DeepHit and DRSA demonstrates an improvement when augmented with the IACL loss, particularly under conditions of extreme event-censoring imbalance. For all models, the parameter γ is consistently set to 2.

In contrast, given that the imbalance issue is less pronounced in the SUPPORT and FLCHAIN datasets, the magnitude of performance enhancement observed is comparatively modest.

Table 5: C^{td} -index performance comparison with variation of γ in the loss function on INSURANCE dataset

Dataset	γ				
	0	0.5	1	2	5
INSURANCE	0.610	0.612	0.613	0.614	0.613

Table 6: C^{td} -index of different models on INSURANCE, SUPPORT and FLCHAIN data sets

Methods	INSURANCE	SUPPORT	FLCHAIN
DeepHit w IACL	0.607	0.719	0.776
DeepHit w/o IACL	0.604	0.719	0.774
DRSA w/ IACL	0.610	0.723	0.779
DRSA w/o IACL	0.607	0.722	0.776
SurvFormer	0.614	0.730	0.789
SurvFormer w/o IACL	0.610	0.730	0.787

5 Conclusions

In conclusion, our study presents SurvFormer, a novel approach to survival analysis that leverages the transformer-decoder architecture alongside a novel imbalance-aware censoring loss (IACL). Our model leverages attention mechanism to emphasize relevant input features and capture the temporal state inter-dependencies along the prediction time window. Additionally, the parallel computation capability ensures that our model maintains high efficiency, making it well-suited for handling large-scale industry datasets. By integrating the IACL, our model effectively mitigates the challenge of event-censoring imbalance commonly encountered in the health insurance data. The empirical results from extensive experiments on the three real world datasets demonstrate that SurvFormer outperforms existing state-of-the-art methods in survival analysis, offering a robust and generalizable tool for predicting event timing with high accuracy, particularly within large-scale industry contexts. For the future work, it is natural to integrate the transformer encoder to the SurvFormer framework to adapt to the sequential input features and also the situations where events may occur repeatedly and have continuous values.

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